

Vanadis 3D vs Fluidity-Atmosphere, MFEM, CMAQ and FLEXPART

Computational architecture comparison

Focus: FEM structure, global matrices, Element-by-Element execution, GPU/CPU parallelism, memory, solvers, stabilization and atmospheric role.

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Scope and disclaimer. This document compares computational and software architecture only. It is not a ranking of scientific accuracy, validation maturity, atmospheric chemistry breadth, operational readiness, or measured performance. The systems serve different purposes: MFEM is a general FEM library, CMAQ is a comprehensive Eulerian chemical transport model, FLEXPART is a Lagrangian particle model, and Fluidity-Atmosphere is an atmospheric research framework. Statements are based on the Vanadis v2026.3.0 project state and publicly available documentation checked on 7 September 2026. Project names are used for technical identification only; no affiliation or endorsement is implied.

Architectural comparison

Feature	Vanadis 3D	Fluidity-Atmosphere	MFEM	CMAQ	FLEXPART
Primary numerical method	3D FEM, HEX8 elements	Mixed FEM/FV on unstructured, adaptive meshes	General-purpose FEM library	Eulerian chemical transport model; grid-based finite-volume transport	Lagrangian particle dispersion model
Global sparse matrix	No conventional global sparse matrix in the EbE solver path	Yes; local contributions are assembled into global CSR/BCSR matrices	Optional: FULL/LEGACY use global sparse matrices; ELEMENT/PARTIAL/NONE can avoid them	Not a FEM matrix architecture; prognostic 3D fields are advanced process by process	No global PDE matrix
Element-by-Element through solver	Yes	No; element work is followed by global sparse assembly	Supported at ELEMENT assembly level; PARTIAL/NONE go further toward matrix-free execution	Not applicable	Not applicable
Stored element matrices	Yes - bb stores element matrices	Local element data/matrices are formed, then assembled globally	Optional at ELEMENT level	No FEM element matrices	No
GPU execution	CUDA EbE linear solver	CUDA element integration and global assembly in the referenced GPU implementation; PETSc solution remains CPU-side there	CUDA, HIP, RAJA, OCCA, libCEED and other backends	No CUDA backend is documented in the standard CMAQ 5.5 CCTM configuration examined; standard execution is CPU/MPI	Standard FLEXPART 11 uses CPU/OpenMP parallelism; GPU execution is not part of the documented standard path
OpenMP	Yes - CPU element assembly	Referenced atmospheric GPU implementation emphasizes MPI + CUDA rather than OpenMP	Yes	Standard CMAQ 5.5 CCTM uses MPI; its configuration explicitly notes that OpenMP is not supported	Yes - FLEXPART 11 uses OpenMP
MPI	Not the basis of the current Vanadis architecture	Yes	Yes	Yes - principal domain parallelism	Earlier FLEXPART versions used MPI; version 11 moved to OpenMP

Feature	Vanadis 3D	Fluidity-Atmosphere	MFEM	CMAQ	FLEXPART
Memory architecture	One large shared bb in CPU RAM plus one device copy in VRAM; small per-thread work arrays	Element data plus connectivity plus global CSR/BCSR matrix and host/device transfer requirements	Depends on assembly level: FULL is memory-heavy; PARTIAL/NONE greatly reduce stored operator data	Many 3D species, meteorological, aerosol and process fields	Particle arrays plus meteorological fields
Memory growth with more CPU threads	Small relative growth: bb is shared; mainly private/threadprivate work data are replicated	Depends strongly on MPI decomposition and GPU/global-matrix representation	Backend and assembly-level dependent	Driven mainly by domain decomposition, species count and model fields	Shared-memory OpenMP; no FEM matrix replication
Linear / algebraic solver	DBCG with diagonal/Jacobi preconditioning	PETSc iterative solvers and preconditioners	Broad solver ecosystem: CG/GMRES/BiCGSTAB/MINRES, Hypr, PETSc, etc.	No single global FEM-style solver; transport and chemistry use specialized algorithms	No global linear PDE solver; particle trajectories are advanced directly
Advection stabilization	Directional Residual (DR)	Uses its own stabilization/discretization mechanisms; not DR	Application-dependent; MFEM provides building blocks rather than one imposed method	PPM-type monotone, positive-definite finite-volume transport	Lagrangian transport avoids the usual Eulerian grid-advection stabilization problem
GPU atomics	Yes - atomic EbE variant	Used for global sparse assembly in the referenced GPU implementation	Operator/backend dependent	Not applicable in this architectural sense	Not applicable
GPU graph coloring	Yes - coloring variant avoids atomic updates within each color	Alternative synchronization strategies can be explored, but the published atmospheric GPU path centers on global assembly	Application/backend dependent	Not applicable	Not applicable
Nonlinearity	Concentration-dependent P(S), solved by Picard iteration	Supports nonlinear atmospheric systems	Extensive nonlinear operator/solver support	Strongly nonlinear atmospheric chemistry	Decay, deposition and particle-process parameterizations; not an Eulerian nonlinear matrix formulation
Time-dependent sources	Generalized Q(t) source handling	Possible within the general model formulation	Application-dependent	Detailed time-dependent emissions inventories	Time-dependent releases and receptors supported
Atmospheric role	Dedicated 3D pollutant transport and dispersion model	Atmospheric dynamics / transport research framework	Not an atmospheric model by itself	Comprehensive Eulerian chemical transport model	Dedicated forward/backward atmospheric transport and dispersion model

1. Vanadis vs Fluidity-Atmosphere

The closest atmospheric comparison is Fluidity-Atmosphere because both systems perform intensive element-level work and can use GPUs. The decisive architectural difference comes after the element calculations. Fluidity-Atmosphere assembles local contributions into conventional global sparse CSR/BCSR matrices, whereas Vanadis keeps the operator in Element-by-Element form through the iterative solver.

In the referenced Fluidity-Atmosphere GPU implementation, element integration and global sparse assembly are accelerated on the GPU, but the assembled matrix is transferred for PETSc-based solution. Vanadis avoids the global sparse assembly step entirely in its EbE path: the CUDA DBCG solver works directly with the stored element matrices. This changes memory traffic, synchronization requirements and the way the solver maps to the GPU.

Vanadis: local HEX8 matrices	stored bb	CUDA EbE MATVEC/MTTVEC	DBCG solution
Fluidity-Atmosphere: local matrices	GPU global assembly	CSR / BCSR matrix	PETSc solution

2. Vanadis vs MFEM

Conceptually, MFEM is very close to the memory and operator philosophy of Vanadis, even though MFEM is a general finite-element library rather than an atmospheric dispersion model. MFEM offers FULL, ELEMENT, PARTIAL and NONE assembly levels. The ELEMENT level stores element matrices instead of a conventional global sparse operator, which is close to the role of bb in Vanadis.

MFEM goes further with partial assembly and matrix-free approaches, especially for high-order methods. These modes reduce memory traffic by storing only the data required to apply the operator. Vanadis is more specialized: it fixes the physical model, HEX8 discretization, DR stabilization and EbE iterative solver, while MFEM supplies a broad toolkit from which many different applications can be built.

3. Vanadis vs CMAQ

CMAQ represents a fundamentally different numerical architecture. It is an Eulerian chemical transport model operating on gridded three-dimensional fields. Transport, chemistry, deposition and other processes are advanced through specialized modules rather than through a global FEM algebraic system.

The dominant memory cost in CMAQ comes from the large number of three-dimensional chemical, aerosol and meteorological fields. Parallelism is based mainly on MPI domain decomposition. Therefore concepts such as element matrices, bb, Element-by-Element matrix-vector products and DBCG do not have direct equivalents in CMAQ.

4. Vanadis vs FLEXPART

FLEXPART is even more different numerically because it is a Lagrangian particle dispersion model. Instead of solving pollutant concentration on a finite-element or finite-volume grid, it follows a large ensemble of computational particles through meteorological fields.

This means that FLEXPART has no global PDE matrix and no FEM Element-by-Element solver. Its memory cost is driven mainly by particle state arrays and meteorological input fields. The Lagrangian formulation also avoids the classical Eulerian grid-advection stabilization problem, so there is no direct analogue of DR.

Memory architecture - dominant cost

System	Dominant memory structure
Vanadis	Stored element matrices bb + nodal vectors + connectivity; bb remains shared across OpenMP threads. GPU uses one device copy of bb.
Fluidity-Atmosphere	Element data + connectivity + assembled global CSR/BCSR operator and host/device data movement.
MFEM	From full sparse matrices to element, partial-assembly or matrix-free data depending on the selected assembly level.
CMAQ	Large collections of 3D species, meteorological, aerosol and process fields.
FLEXPART	Particle arrays plus meteorological fields; no FEM/global sparse operator.

What is distinctive about Vanadis

No single Vanadis feature is unique in isolation. FEM, GPU acceleration, element-local operators, Krylov solvers and atmospheric dispersion all exist elsewhere. What is unusual is their combination in one dedicated atmospheric model: 3D FEM + Directional Residual stabilization + stored element matrices + no conventional global sparse matrix + Element-by-Element iterative solution + OpenMP CPU assembly + CUDA DBCG + atomic and graph-coloring GPU variants + nonlinear Picard P(S) + time-dependent Q(t). The especially important design choice is that Vanadis preserves the Element-by-Element structure all the way through the solver rather than returning to a global sparse matrix after local element calculations.

References and source basis

- [1] Vanadis 3D project repository and releases (v2026.3.0 used as the Vanadis reference state): https://github.com/marek-ac/vanadis_solver-directional_residual
- [2] Fluidity-Atmosphere GPU implementation: GPU-accelerated finite-element method for the three-dimensional unstructured mesh atmospheric dynamic framework, Geoscientific Model Development 19 (2026): <https://gmd.copernicus.org/articles/19/7525/2026/>
- [3] MFEM documentation - Assembly Levels (FULL, ELEMENT, PARTIAL, NONE): https://mfem.org/howto/assembly_levels/
- [4] MFEM documentation - performance and partial assembly / GPU-oriented operator application: <https://mfem.org/performance/>
- [5] U.S. EPA CMAQ v5.5 source and User Guide: <https://github.com/USEPA/CMAQ>
- [6] CMAQ v5.5 model configuration - PPM transport; config_cmaq.csh - MPI toolchain and OpenMP note: https://github.com/USEPA/CMAQ/blob/main/DOCS/Users_Guide/CMAQ_UG_ch06_model_configuration_options.md ; https://github.com/USEPA/CMAQ/blob/main/config_cmaq.csh
- [7] FLEXPART official project description: <https://www.flexpart.eu/>
- [8] FLEXPART roadmap - version 11 OpenMP parallelisation: <https://www.flexpart.eu/roadmaps.html>

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